Elements of AIML
Miscellaneous – Lecture 02 Notes
Eigenvalues and Eigenvectors from a Data Sample

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1 Lecture Overview

This supplementary lecture shows the full manual path from a tiny **raw data sample** to:

- the mean vector,
- the centered data matrix,
- the covariance matrix,
- the eigenvalues,
- the eigenvectors,
- the PCA interpretation of those eigenvectors.

This is useful because students often understand the definitions but do not see how the calculation is chained from the original sample.

Sources: Jolliffe and Cadima, "Principal component analysis: a review and recent developments," *Philosophical Transactions of the Royal Society A*, 2016.;Fisher, "The use of multiple measurements in taxonomic problems," *Annals of Eugenics*, 1936.;Hyvärinen and Oja, "Independent component analysis: algorithms and applications," *Neural Networks*, 2000.;Bishop, *Pattern Recognition and Machine Learning*, selected sections on PCA, LDA, and ICA.;Scikit-learn documentation, accessed 2026-04-18.

2 How to use these notes

Use this lecture after:

- **Unit 03 Lecture 03**, where PCA is introduced conceptually,
- **Miscellaneous Lecture 01**, if you want to know why PCA and LDA are tied to eigenanalysis.

The present lecture is not mainly about comparing methods; it is about doing one derivation carefully and correctly.

3 Learning Outcomes

- Compute the mean vector from a small dataset.
- Center the data correctly.
- Build the covariance matrix from the centered data.
- Solve the characteristic equation to obtain eigenvalues.
- Solve the linear system for eigenvectors.
- Interpret the largest-eigenvalue direction as the first principal component.

4 Raw sample data

Consider the following three data points:

$$x_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \qquad x_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad x_3 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

These are deliberately small so the entire calculation can be done by hand.

5 Step 1: Compute the mean vector

The sample mean is:

$$\mu = \frac{1}{3}(x_1 + x_2 + x_3)$$

Substitute the values:

$$\mu = \frac{1}{3}\left(\begin{bmatrix}2\\1\end{bmatrix} + \begin{bmatrix}1\\2\end{bmatrix} + \begin{bmatrix}0\\0\end{bmatrix}\right) = \frac{1}{3}\begin{bmatrix}3\\3\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix}$$

6 Step 2: Center each sample

Centering means subtracting the mean vector from every sample:

$$x_i^{(c)} = x_i - \mu$$

So:

$$x_1^{(c)} = \begin{bmatrix}2\\1\end{bmatrix} - \begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}1\\0\end{bmatrix}$$

$$x_2^{(c)} = \begin{bmatrix}1\\2\end{bmatrix} - \begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}0\\1\end{bmatrix}$$

$$x_3^{(c)} = \begin{bmatrix}0\\0\end{bmatrix} - \begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}-1\\-1\end{bmatrix}$$

Therefore the centered data matrix is:

$$X_c = \begin{bmatrix}1 & 0\\0 & 1\\-1 & -1\end{bmatrix}$$

where each row is one centered sample.

7 Step 3: Compute the covariance matrix

For $n = 3$ samples, the sample covariance matrix is:

$$C = \frac{1}{n-1}X_c^\top X_c = \frac{1}{2}X_c^\top X_c$$

7.1 Compute $X_c^\top X_c$

$$X_c^\top = \begin{bmatrix}1 & 0 & -1\\0 & 1 & -1\end{bmatrix}$$

Now multiply:

$$X_c^\top X_c = \begin{bmatrix}1 & 0 & -1\\0 & 1 & -1\end{bmatrix}\begin{bmatrix}1 & 0\\0 & 1\\-1 & -1\end{bmatrix}$$

$$= \begin{bmatrix}1^2 + 0^2 + (-1)^2 & 1\cdot 0 + 0\cdot 1 + (-1)(-1)\\0\cdot 1 + 1\cdot 0 + (-1)(-1) & 0^2 + 1^2 + (-1)^2\end{bmatrix} = \begin{bmatrix}2 & 1\\1 & 2\end{bmatrix}$$

So:

$$C = \frac{1}{2}\begin{bmatrix}2 & 1\\1 & 2\end{bmatrix} = \begin{bmatrix}1 & 0.5\\0.5 & 1\end{bmatrix}$$

8 Step 4: Find the eigenvalues

We solve:

$$\det(C - \lambda I) = 0$$

Substitute C:

$$\det \begin{bmatrix} 1-\lambda & 0.5 \\ 0.5 & 1-\lambda \end{bmatrix} = 0$$

So:

$$(1-\lambda)^2 - (0.5)(0.5) = 0$$

$$(1-\lambda)^2 - 0.25 = 0$$

$$1 - 2\lambda + \lambda^2 - 0.25 = 0$$

$$\lambda^2 - 2\lambda + 0.75 = 0$$

Factor:

$$(\lambda - 1.5)(\lambda - 0.5) = 0$$

Hence the eigenvalues are:

$$\lambda_1 = 1.5, \qquad \lambda_2 = 0.5$$

9 Step 5: Find the eigenvectors

9.1 Eigenvector for $\lambda_1 = 1.5$

Compute:

$$C - 1.5I = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix} - \begin{bmatrix} 1.5 & 0 \\ 0 & 1.5 \end{bmatrix} = \begin{bmatrix} -0.5 & 0.5 \\ 0.5 & -0.5 \end{bmatrix}$$

Now solve:

$$\begin{bmatrix} -0.5 & 0.5 \\ 0.5 & -0.5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

This gives:

$$-0.5x + 0.5y = 0 \Rightarrow y = x$$

So one eigenvector is:

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

9.2 Eigenvector for $\lambda_2 = 0.5$

Compute:

$$C - 0.5I = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$$

Now solve:

$$\begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

This gives:

$$0.5x + 0.5y = 0 \Rightarrow y = -x$$

So one eigenvector is:

$$v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

10 Step 6: Normalize the eigenvectors

To convert them into unit vectors:

$$\|v_1\| = \sqrt{1^2 + 1^2} = \sqrt{2}, \qquad \|v_2\| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$

Therefore:

$$\hat{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \hat{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

11 Interpretation

Because:

$$\lambda_1 = 1.5 > \lambda_2 = 0.5$$

the direction:

$$\hat{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

captures more variance than the perpendicular direction.

11.1 PCA meaning

If we use this covariance matrix in PCA:

- $\hat{v}_1$ becomes the first principal component,
- $\hat{v}_2$ becomes the second principal component,
- the first component is more informative because its eigenvalue is larger.

12 What this example teaches

This worked example shows a full chain:

1. start from raw samples,
2. compute the mean,
3. center the samples,
4. build the covariance matrix,
5. solve the characteristic equation,
6. solve for the eigenvectors,
7. interpret the result in PCA language.

13 Common mistakes

- **Skipping centering:** then the covariance matrix does not reflect spread around the mean correctly.
- **Using the raw data matrix directly:** in PCA the eigen-problem is applied to the covariance matrix, not to the raw sample matrix.

- **Ignoring normalization:** direction is enough mathematically, but unit vectors make the interpretation cleaner.
- **Confusing largest eigenvalue with largest coordinate:** the eigenvalue measures variance along the direction, not the magnitude of one coordinate.

14 Practice Questions (with answers)

Q1. Why do we divide by $n - 1$ in the covariance formula?

Answer: because we are computing the sample covariance, which uses Bessel's correction.

Q2. Why is the covariance matrix symmetric?

Answer: because covariance between feature 1 and feature 2 equals covariance between feature 2 and feature 1, so $C = C^\top$.

Q3. Which eigenvector becomes PC1?

Answer: the eigenvector corresponding to the largest eigenvalue.

Q4. Why is this example useful for PCA but not enough for ICA?

Answer: because PCA depends directly on covariance eigen-analysis, while ICA requires a stronger independence-based optimization beyond covariance alone.

15 Summary

For a tiny dataset, eigenvalues and eigenvectors can be found by hand. This derivation is important because it makes the PCA procedure less mysterious:

- covariance summarizes feature spread,
- eigenvectors give the principal directions,
- eigenvalues tell how much variance those directions capture.

Sources: Jolliffe and Cadima, "Principal component analysis: a review and recent developments," *Philosophical Transactions of the Royal Society A*, 2016.;Fisher, "The use of multiple measurements in taxonomic problems," *Annals of Eugenics*, 1936.;Hyvärinen and Oja, "Independent component analysis: algorithms and applications," *Neural Networks*, 2000.;Bishop, *Pattern Recognition and Machine Learning*, selected sections on PCA, LDA, and ICA.;Scikit-learn documentation, accessed 2026-04-18.